Multirate Progressive Entropy Model for Learned Image Compression

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***Abstract*— This paper proposes a unified and efficient entropy coding method for learned image compression (LIC) from the perspective of traditional signal processing. First, the consistency of structures and optimization objectives are used to interpret the existing split-coded-then-merge entropy coding strategies in LIC as a particular filter banks framework, with feature separation and feature aggregation representing the analysis filter bank and synthesis filter bank, respectively. Thus, we borrow the design from the multirate filter banks and proposed Multirate Progressive Entropy Model (MPEM) to enhance the rate-distortion performance and decoding speed. In particular, we create an analysis filter bank that divides compact features into a few nonuniform subsets based on various spatial and channel sampling rates. Then multi-scale detail and mean coefficients within the current subset are used as prior representations to help generate the prediction parameters of the next subset, and the carefully designed synthetic filter bank performs a near-perfect reconstruction of the features. In addition, we propose a Multi-level Edge Attention Moudal (MEAM) to increase the edge and texture information's contribution and reduce the high-frequency information loss brought on by MPEM's inherent multi-rate spatial sampling, which leverages the edge operator and structural reparameterization principles. The results of the experiments show that, in comparison to the effective LIC methods and traditional code, the proposed MPEM can decode data at a cutting-edge speed while also offering comparable rate-distortion performance.**

***Index Terms*— Deep learning, image compression, multi-rate filter bank, entropy model, attention mechanism.**

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I. Introduction

THE rapid growth of multimedia data necessitates the development of efficient mass data transmission and storage methods. Hence, lossy image compression has become one of the crucial and fundamental technologies in the multimedia community. Lossy image compression aims to token the image within the limit of bit and maintain the quality of the image. Toward this end, many traditional hybrid image compression frameworks have been proposed and improved continuously, including JPEG [1], JPEG2000 [2], HEVC [3], AVS [4] and VVC [5]. However, these methods are typically constructed by independent hand-design modules. As a result, it is challenging to jointly optimize the entire framework, and easy to get stuck at rate-distortion (R-D) performance that is locally optimal [6]. The learned image compression framework is emerging due to this trend, which incorporates learnable network structure into the end-to-end training process and aims to optimize the compression frame globally.

Learned image compression (LIC) techniques have significantly progressed in recent years, outperforming current hybrid image codecs and R-D benchmarks. In addition, the entropy model plays a crucial role in estimating the conditional probability distribution of latent variables, as demonstrated by studies [7], [8], [9] into the interpretability and robustness of the black-box system of LIC. Better performance with fewer bits can be achieved by incorporating a robust and precise entropy model into LIC. As a result, the advanced design of the entropy model has emerged as a significant area of LIC research.

Ballé et al. [10] are credited as the pioneers of proposing a factorized entropy model and performing end-to-end rate-distortion optimization. Hyperprior entropy coding [11] and its variants [12], [13] use zero-mean Gaussian and other advanced probability models to perform hyperprior prediction, building on this foundation. In addition, several auxiliary context [14], [15], [16], [17], [18], [19], [20] models have been added as optional extensions of hyperprior prediction to reduce compression rates and improve estimation accuracy. However, it should be noted that most context models come at the expense of decoding efficiency, as sequential spatial autoregressive strategies tend to disrupt parallelism significantly. Consequently, many researchers are now exploring designs for more efficient entropy models.

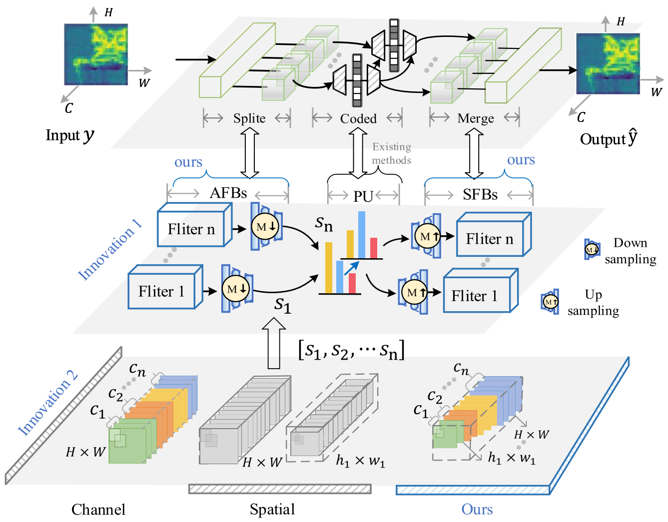


Fig. 1. Illustration of the consistency between split-coded-then-merge entropy model and filter banks. The top row represents split-coded-then-merge strategy in entropy model. The second row represents traditional filter banks, where AFBs, PU, SFBs, M denote the analysis filter banks, the processing unit, the synthetic filter banks and sampler respectively, and $[s_1, s_2, \cdots, s_n]$ stands for the lower-dimensional embeddings after AFBs. We consider the split-coded-then-merge entropy model as special case of filter banks.

Adopting a channel-conditioning context model [21] instead of a serial-decoded spatial model, which makes parallel decoding easier, is one viable strategy for combating parallelism corruption. Enhancing estimation [22], [23], [24] or lightweight spatial context models is yet another widely used strategy. One notable example is checkerboard context model [25], which achieves parallel context modeling among spatial neighbors by employing a two-step checkerboard pattern. Uneven space-channel context model [26] and multistage context model [27] have recently combined the uneven grouping strategy with checkerboard convolution. Therefore, impressive R-D performance can be achieved in a shorter decoding time. As depicted in the top row of Fig. 1, a split-coded-then-merge strategy is a significant property that all of the aforementioned entropy models generally share. The input is divided into a few lower-dimensional embeddings in an entropy model, processed by specialized regression techniques, and combined by concatenation. However, most of the split-coded-then-merge entropy model designs are unsystematic, and great exploration potential still exists to improve the decoding speed further and reduce computation complexity.

In contrast to existing methods, this paper provides a novel viewpoint on the split-coded-then-merge entropy model from the perspective of traditional filter banks [28], [29]. It shows how this idea can be extended to systematically design an efficient entropy model. First, the existing split-coded-then-merge entropy coding strategies are interpreted as a particular filter banks framework according to the consistency in structures and optimization objectives, where feature separation, auto-regression, and feature aggregation correspond to analysis filter bank, processing unit and synthesis filter bank, respectively. Fig. 1 specifies the mappings between generalized filter banks and split-coded-then-merge entropy models. Then, the design concept of filter banks are used to examine two types of common split-process-then-merge LIC entropy models.

Specifically, most entropy models convert the output after transforming networks into lower-dimensional embeddings that facilitate rapid coding. In the meantime, the currently used methods can be divided into two groups based on various dimensions. The lower-dimensional embeddings are, for instance, defined by channel-based methods as several channel subsets arranged in particular ways (such as channel clustering [30] and channel grouping [26], [27], [31]). On the other hand, the spatial-based methods [22], [23], [32] construct lower-dimensional embeddings as several spatial subsets with different resolutions. These are designed to achieve an in-depth analysis of the latent representation and boost R-D performance. However, the lower-dimensional embeddings are generated only along individual dimensions, whether channel-based or spatial-based methods. Hence, great exploration potential still exists to improve the decoding speed further and reduce model complexity. In the meantime, most of approaches concentrate on altering the processing unit to balance R-D performance and decoding speed. In a whole filter banks framework, designing a reasonable and reliable analysis and synthesis filter bank is the key to ensuring the system's perfect or nearly perfect reconstruction [33].

We propose the Multirate Progressive Entropy Model (MPEM) based on the above analysis to improve R-D performance and decoding speed further. Approaching from the filter banks perspective, we emphasize the design of analysis and synthesis filter banks, contrasting with the emphasis placed by most existing methods complex processing unit, as demonstrated in the middle row of Fig. 1. Specifically, we design an analysis filter bank to combine the advantages of the two above grouping operations, which divides compact features into some nonuniform feature subsets according to different channel sampling rates and spatial sampling rates (as depicted in the bottom row of Fig. 1). The prediction parameters for the subsequent subset are then generated using multi-scale detail and mean coefficients from the current subset as prior representations. In addition, a synthesis filter is introduced to perform a nearly perfect reconstruction of sampled subbands, enhancing the R-D performance. Afterward, for the key information loss resulting from the spatial sampling operations in MPEM, a Multi-level Edge Attention Moudal (MEAM) is proposed for the total codec to build a more compact and more discriminating feature space by making full use of edge and texture information. The following is a summary of the contributions made in this paper.

- We model the existing split-coded-then-merge entropy models as generalized filter banks according to the consistency instructions and optimization process, which guides the design of the efficient entropy model.
- A novel entropy approach called Multirate Progressive Entropy Model (MPEM) introduces multirate filter banks into the entropy model design and has a progressive autoregressive process. Due to its fast decoding speed and competitive R-D preference, MPEM is especially well-suited for practical deployment.

- We propose an efficient Multi-level Edge Attention Module that is plug-and-play compatible with any LIC framework. It can be used in conjunction with MPEM to enhance RD performance further to circumvent the limitations of MPEM's high-frequency information loss.

The remainder of this paper is organized as follows: Section II briefly introduces related works. Section III models the problem formulation and then introduces the proposed methods. Section IV gives comparative experiments on performance and efficiency indicators. Section V provides detailed information on the ablation studies and performance analysis. In Section VI, the conclusion and future work are discussed.

II. Related Work

This section reviews related works from two aspects. The first one discourses upon principles for LIC, an emerging topic in recent years. The second is efficient entropy coding, which plays a vital role in practical LIC.

A. Principles for Learned Image Compression

Recent research in LIC has focused on leveraging deep neural networks' powerful nonlinear transformation capabilities to decorrelate signals, followed by building probability models for accurate entropy coding. In common metrics like the multi-scale structural similarity index (MS-SSIM) and peak signal-to-noise ratio (PSNR), this method has demonstrated superiority and potential over traditional methods. A pipeline that includes transformation, entropy coding, and other techniques has been the primary focus of the majority of correlational research conducted in recent years.

1) Transformation: The transforms minimize feature redundancy and thus obtain compact feature representations. The variational autoencoder (VAE)-based structure is recognized as the most successful in modeling such a nonlinear transformation network and has a wide range of applications. Ballé et al. [10], [11] constructed a transformation network using a convolution layer and generalized division normalization (GDN), which served as the basis for subsequent research in LIC. This idea's extensions, such as residual structures and larger networks, have been proposed in works like [6], [34], [35], [36], [37], [38], [39], [40], and [41]. However, comprehensive transform networks with excellent R-D performance often require high computational complexity, slowing the overall inference.

The GDN variants [42], [43], [44], in contrast to the comprehensive structural design, can be easily incorporated into the transformation network as modular components. As a result, improving the R-D performance of the network is more facilitated. Moreover, by incorporating a non-local module into LIC, the residual non-local image compression [6], [34], [45], the model eliminates local and global correlations between pixels. Zuo et al. [46] proposes a window-based attention module that flexibly captures correlations among spatially neighboring elements. Therefore, LIC's capacity for nonlinear transformation is further enhanced. However, the majority of these attention methods suffer from certain limitations. Although the modular design may help mitigate the significant increase in overall complexity, the attention methods usually rely on the non-local operation. As a result, high computational complexity is still required for the whole nonlinear transformation. Meanwhile, while such attention methods effectively carry out local and global correlations, they seldom touch the distinctive importance of texture details and edge information. However, texture details and edge information can be of crucial importance in a wide variety of low-level tasks [47].

2) Entropy Coding: LIC techniques rely heavily on entropy modeling to estimate the codes' rate. The factorized entropy model [10], a pioneering work, is widely used in LIC. However, this model can be susceptible to local optima in the presence of spatial redundancy in the latent representation. To overcome this issue, hyperprior entropy coding [11] and its variants [12], [13], [48] utilize the zero-mean Gaussian and other advanced probability models to eliminate spatial redundancy and improve the R-D performance of LIC. Subsequently, the joint autoregressive [14] reduces spatial redundancy by using side information and neighboring elements in latents. Numerous auxiliary context models [16], [34], [35], [49], [50], [51]—such as the global context model [50] and the causal context model [16]—have been proposed recently to take advantage of latent correlations to enhance bit reduction and accurately model entropy. However, the above context models come at the cost of decoding efficiency. A more fundamental explanation is that spatial autoregressive strategy significantly breaks the parallel. Hence, how to effectively enhance the decoding efficiency of entropy coding has become a research hot topic in recent years.

B. Efficient Entropy Coding

Accurate entropy coding has a wide range of applications in LIC, but most still suffer from slow-speed issues. Many efficient grouping schemes have recently been introduced to entropy coding to speed up the decoding. Those existing methods can be divided into two categories according to different processing dimensions of feature representation.

1) Channel-Based Methods: Minnen et al. [21] replace the operation in spatial dimensions with channel even grouping in channel-conditioning models to improve parallelism. Following the energy compaction in LIC, uneven space-channel context model [26] and multistage context model [27] the following improvement works to expand this even grouping to uneven grouping and then combine it with checkerboard context models [25]. Therefore, decoding time can be reduced while R-D performance is improved. Yuan et al. [30] present channel-wise grouping based on a clustering algorithm to exploit the channel-wise correlations further. It develops it as a grouped 3-D context model, which obtains efficient entropy coding with a considerable R-D performance. Although channel grouping operations can improve processing speed, rough grouping schemes may undermine the channel-wise correlation, resulting in R-D performance degradation. Moreover, this is why the channel-based method often requires additional auxiliary modules (e.g., latent residual prediction and checkerboard context models).

2) Spatial-Based Methods: Hu et al. [22] construct a multi-scale coarse-to-fine hyperprior model to guide the learning of latent representations for fully exploiting the potential of spatial dependencies in the latent representation. Fu et al. [23] utilize octave convolution and an additional hyper layer instead of the context-adaptive model to eliminate the latent representation's spatial redundancy, leading to a better tradeoff between R-D performance and decoding speed. However, the sampling operations of inter-frequency communication destroy the scale information of the latent representation, which may limit the performance of LIC. In this regard, a context transfer module is added to two-stage [32] to capture the redundancy between the multi-frequency latents. However, it also led to massive computational complexity, inevitably slowing the decoding speed.

Differing from the above two methods, we model the existing split-coded-then-merge entropy model as a generalized filter banks according to the consistency instruction and optimization process. Then, we design an efficient entropy model from the perspective of a multirate filter banks [52]. As the advantages of channel-based and spatial-based technologies were combined, they could adapt to rapid LIC decoding.

III. Proposed Method

From a filter banks perspective, we first show how to analyze and model the entropy coding and highlight the current limitations. Then, the MPEM is proposed to obtain an acceptable tradeoff between nearly perfect reconstruction (NPR) of latent representation and decoding speed. Finally, we propose a multi-level edge attention module to mitigate the high-frequency information loss due to MPEM.

A. Problem Formulation

As shown in Fig. 1, for the transformed compact feature representation $y \in \mathbb{R}^{C\times H\times W}$, three steps make up the split-coded-then-merge entropy model: 1) Feature decomposition and sampling along spatial or channel dimension over the compact feature representation y; 2) Encoding and decoding the sampled subsets with a specific probability estimate provided by the processing unit; 3) Reconstitution and feature synthesis using the processed feature subsets.

Specifically, they first decompose the compact feature representation y into n feature subsets $S = \{s_1, s_2, \ldots, s_n\}$, which can be determined as follows:

$$S = A_f(\mathbf{y}; \phi), \tag{1}$$

where $A_f(\cdot)$ is a general expression that adheres to certain rules (such as artificial experience, clustering algorithm, and additional hyperlayer), and ϕ is a learnable parameter. Table I summarizes the existing split-coded-then-merge entropy model in LIC based on the difference in sampling direction and rate.

For instance, the feature subset frequently takes the form of $s_n \in \mathbb{R}^{c_n\times H\times W}(c_n < C)$ when the method employs filtering and sampling along channel direction over the compact feature representation. By contrast, the compact feature representation always comes out with the $s_n \in \mathbb{R}^{C\times h_n\times w_n}(h_n < H, w_n < W)$ when the decomposition is loaded along the constant spatial dimension. Moreover, When all feature subsets contain two or more sampling rates $c_n \neq c_{n-1}$ or $h_n, w_n \neq h_{n-1}, w_{n-1}$,

TABLE I

The Existing Split-Coded-Then-Merge Entropy Model Are Summarized According to the Difference of Sampling Direction and Rate. Ch Is Defined as the Channel Dimension, and Sp Denotes the Spatial Dimension

Methods	$A_f(\cdot)$	s_n		Sample Rate	$P_u(\cdot)$	$G_f(\cdot)$
		ch	sp			
Minnen2020 [21]	-	✔	✗	single-rate	ch context	-
Yuan2021 [30]	Cluster	✔	✗	multi-rate	ch context	Recover
He2022[26]	-	✔	✗	multi-rate	ch&sp context	-
Lu2022 [27]	-	✔	✗	multi-rate	ch&sp context	-
Hu2020 [22]	-	✗	✔	single-rate	sp prediction	-
Fu2022 [23]	-	✗	✔	single-rate	sp prediction	-
MPEM(ours)	Synthetic filter bank	✔	✔	multi-rate	ch context& sp prediction	Analysis filter bank

this process is called multi-rate sampling. Single-rate sampling is when all sampling rates are the same. Since multi-rate sampling makes flexible and effective use of the dynamic signal characteristic, it brings more benefits like less memory and high processing speed than single-rate sampling, which is also confirmed in [26] and [27].

The sampled feature subsets are then imputed into the processing unit for further compression and probability estimation. The corresponding formulation is given by:

$$\hat{S} = \left\{\hat{s}_1, \hat{s}_2, \ldots, \hat{s}_n\right\} = P_u(\mathbf{S}), \tag{2}$$

where $P_u(\cdot)$ is defined as a processing unit, including but not limited to channel condition, checkerboard, and other advanced context models. $\hat{S} = \left\{\hat{s}_1, \hat{s}_2, \ldots, \hat{s}_n\right\}$ represents the processed feature subsets. The signal synthesis can now be formulated in the following manner:

$$\hat{y} = G_f(\hat{\mathbf{S}}; \theta), \tag{3}$$

where $G_f(\cdot)$ is a generalized signal synthesis with learnable parameters θ, and $\hat{y}$ represents the final reconstruction compact representation.Thus, the whole process of split-coded-then-merge entropy coding can be combined into the following formula:

$$\hat{y} = G_f(P_u(A_f(\mathbf{y}; \phi)); \theta). \tag{4}$$

In summary, the above split-coded-then-merge of entropy model is essentially a subband coding or Filter Banks technology [28], [29]. Moreover, hence we reconsider Eq.(4) from a Filter Banks perspective. We notice that the $A_f(\cdot)$ can be regarded as a series combination of filtering and sampling, called analysis filter bank in signal processing, which aims to achieve efficient signal decomposition to accelerate the process while maintaining basic signal reconstruction conditions simultaneously. Moreover, an uneven grouping of entropy models [26], [27], [30] also embodies the Multirate Filter Banks [33]. In fact, and $P_u(\cdot)$ of the entropy model can be directly mapped into the processing unit in the filter banks. After that, the reconstruction process $G_f(\cdot)$ is regarded as a special example of a synthesis filter bank. The two methods' ultimate objectives are reducing various distortions and improving recovery of the decomposed signal. From consistency in structures and optimization objectives, the filter banks in signal processing can unify the split-coded-then-merge of the LIC entropy model based on the aforementioned

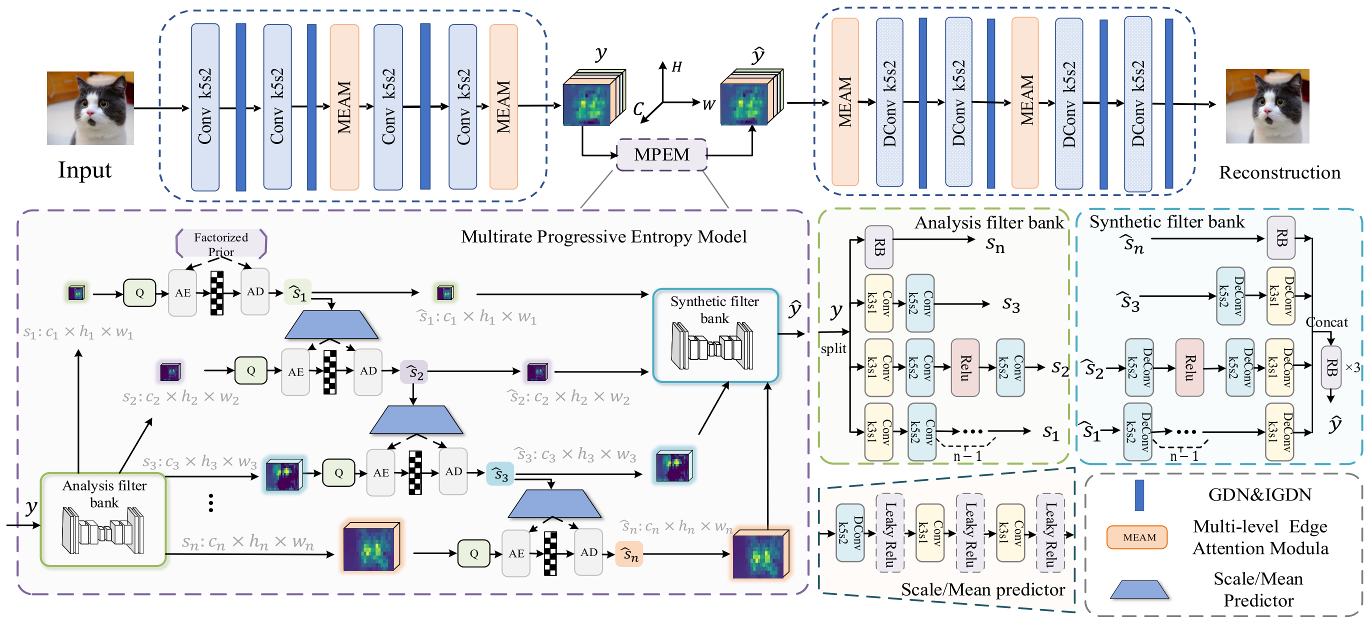


Fig. 2. Illustration of our methods on ICLR2018 [11] network architecture, where the convolution parameters are described as k denotes the size of the convolution kernel, and s denotes the stride. Q represents quantization, and AE and AD represent the arithmetic encoder and arithmetic decoder. Moreover, Multirate Progressive Entropy Model will be described in detail in section III-C.

analysis. Thus, we summarize the limitations that exist in the current study according to the design philosophy of filter banks as follows:

- The shape of feature subsets S suggests that existing methods only group or sample along individual dimensions. Hence, great exploration potential still exists to improve the decoding speed further and reduce computation complexity.
- From the perspective of $P_u(\cdot)$, most methods introduce an advanced context model to reduce spatial correlations further; hence, additional complexity is inevitably introduced in the entropy modeling process.
- In addition, from Eq.(4), the most existing methods focus on modifying the $P_u(\cdot)$ to enhance R-D performance. However, for filter banks, the reliable design of analysis filter banks $A_f(\cdot)$ and synthesis filter banks $G_f(\cdot)$ is one of the core technologies for realizing nearly perfect reconstruction.

To attack these problems, we can refer to the improved filter banks techniques for guiding the design of an efficient entropy model. As a result, in Section III-B, a brand-new entropy model named the Multirate Progressive Entropy Model (MPEM) is developed based on the idea of multirate filter banks [33].

B. Multirate Progressive Entropy Model

Although R-D performance can be significantly improved by combining hyperpriors and spatial neighbors for context modeling, this substantial increase in decoding time resulting from the sequential processing of autoregressive estimation is unacceptable in many efficiency-critical cases. Thus, inspired by a multirate filter banks [33], we proposed a MPEM without the spatial context and extra hyperprior layers, whose key is transforming the emphasis in entropy coding from the complex processing unit to the design of analysis/synthesis filter bank. Such a design greatly simplifies the processing unit and assures the effective utilization of the strong correlation among spatial-channel neighbors.

Based on the multirate filter banks, the proposed MPEM with a three-step procedure: 1) Analysis filter bank subband modeling, 2) Accurate estimation via a lightweight processing unit, and 3) Reconstruction by the synthetic filter bank. The three steps of this procedure are depicted in Fig 2.

Step 1 (Analysis Filter Bank Subband Modeling): The analysis filter bank is composed of several branches. It is crucial to note that in end-to-end image compression, filter building can make it difficult to achieve flawless feature representation reconstruction. Therefore, to analyze filter banks, we opt for serial combinations of multi-branch convolutional filters and samplers (as depicted on the right in Fig 2). When multi-rate filter banks theory is applied to communication and subband coding of speech, there are numerous advantages in processing complex signals, including high resolution and processing speed. Hence, we emphasize the parallel processing and adopt the multirate sampling to further mine the potential value of spatial&channel dimension for following high-efficiency coding. The process is as follows, using the four-branch analysis filter bank as an example:

- We first sample the compact feature representation $y \in \mathbb{R}^{C\times H\times W}$ into four subsets $\{s_1^{'}, s_2^{'}, s_3^{'}, s_4^{'} \in \mathbb{R}^{\{c_1,c_2,c_3,c4\}\times H\times W}$ along the channel dimension, which also provides a set of incremental sampling rates.
- After channel sampling, the subsets will be sent to the appropriate branch for filtering and spatial sampling. Similar to channel sampling, spatial sampling manifests in incremental multirate form. At last, we obtain a fully sampled subset $\{s_1, s_2, s_3, s_4\} \in$

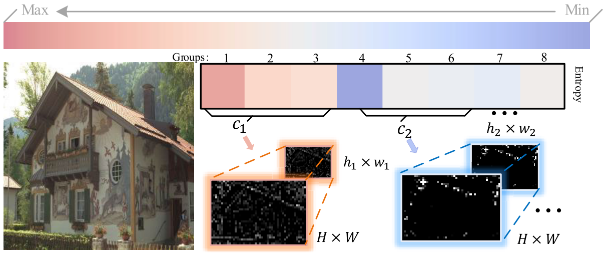


Fig. 3. A case study using channel conditional coding [21]. The entropy of each group of channels is implicitly ordered, with the earliest encoded group containing the channel with the largest entropy. These results were obtained using a grouping model trained with $\lambda = 0.0483$ and evaluated on $kodat24$.

$\mathbb{R}^{\{c_1,c_2,c_3,c_4\}\times\{h_1,h_2,h_3,h_4\}\times\{w_1,w_2,w_3,w_4\}}$.So far, an analysis filter bank subband modeling has been completed.

On the other hand, the proposed analysis filter banks aligns well with the energy concentration property of LIC. As depicted in Fig. 3, this property is frequently represented as a group-level order in the channel-based entropy models [21], [26]. Specifically, the entropy of the earlier encoded channel groups tends to be higher, indicating that more spatial information is available, while the entropy of the later encoded channel groups tends to be lower due to limited information.

Thus, we can use a larger spatial sampling rate to accelerate processing for the earlier encoded groups (c_1), even if some information is lost in the sampling process. The subsequent probability estimation can withstand this information loss, and the resulting estimation error is negligible. However, for the later encoded channel groups (c_2), which contains only a small amount of information, we use a smaller sampling rate to avoid excessive information loss ($h_2 \times w_2 > h_1 \times w_1$), as it significantly impacts the probability estimation. Further feasibility analysis and results are presented in Section V-A3,A4.

Step 2 (Accurate Estimation Via Lightweight Processing Unit):

As a secondary step, the lightweight processing unit aims to execute the accurate parameter estimation with the minimum computation cost. To this end, we jointly employ channel condition and lightweight scale/mean predictor rather than spatial context to remove the spatial redundancy. Specifically, we first generate the feature probabilities of s_1 using the factorized entropy model to achieve en/de-coding of s_1. Then we obtain an entropy parameter $\psi_1 = (\mu_1, \sigma_1)$ by carrying out scale/mean predictor. The lower right of Fig. 2 depicts the scale/mean predictor. Moreover, we borrow the channel-condition model proposed in [21], and the entropy parameter ψ_1 is used to infer the feature probabilities in s_2. This process can be given by:

$$\psi_2 = g_{cc}(g_{sm}(\psi_1; \theta)), \tag{5}$$

where $g_{sm}(\cdot)$ denotes the Scale/Mean Predictor with learnable parameters; $g_{cc}(\cdot)$ is a channel-condition model. Similarly, we repeat the above steps for the remaining subsets until all feature probability estimations are carried out. Thus, the whole process processing unit can be represented by:

$$\psi_n = g_{cc}(g_{sm}(\psi_{n-1}; \theta_n)), n <= 4, \tag{6}$$

all $g_{cc}(\cdot)$ is set to the same architecture, and each has different learnable parameters due to the difference in channel configuration and spatial resolution. After fully encoding and decoding, we finally obtained the subsets $\{\hat{s}_1, \hat{s}_2, \hat{s}_3, \hat{s}_4\}$.

After setting up step 1, we leverage the channel grouping of the large space sampling rate as a prior to estimate the probability of channel grouping of the small space sampling rate. The major benefit of this approach is that the scale/mean predictor as a learnable upsampler. In previous studies of frequency analysis [53], the upsampler often duplicates the strong frequency components of the input to generate the spectrum of upper layers (as illustrated in Fig. 10). This is equivalent to providing more a priori information for the next entropy encoding/decoding, which is critical for later channel groups with less information. Additionally, the scale/mean predictor extends both the channel and spatial dimensions, enabling us to better utilize the channel and spatial correlations. Further details and results are presented in Section V-A5.

Step 3 (Reconstruction By Synthetic Filter Bank):

After processing the unit, we achieve a nearly perfect reconstruction of feature subsets by the synthetic filter bank. All subsets were first placed in a synthetic filter bank with n branches. Then for the n th branch, the corresponding subset has performed a series of upsampling operations to convert itself $\hat{s}_n \in \mathbb{R}^{c_n \times h_n \times w_n}$ into $\hat{s}_n \in \mathbb{R}^{c_n \times H \times W}$. Subsequently, we utilize a series of convolution filters to achieve further subset reconstruction and finally aggregate all subsets in the channel dimension ($\mathbb{R}^{c_n \times H \times W} \rightarrow \mathbb{R}^{C \times H \times W}$). It is important to note that the analysis and synthetic filter banks are made up of imperfect filters, meaning their inherent nonlinear characteristics would lead to signal distortion. Hence, we input it into a simplified three-layer ResBlock after signal aggregation to increase the reliability and validity of the reconstructed feature.

Advantages: Compared to the present split-coded-then-merge entropy models, the advantages of the proposed MPEM are three-fold:

- Using the analysis filter bank shown in Step 1 allows each feature subset to model multirate inter-channel correlation and adaptively consider its spatial scale information as a grouping criterion in the responses from the original scale space, which is a significant improvement over previous grouping procedures.
- The proposed processing unit's structure is straightforward, and its processing speed is superior to that of the complex spatial context model. Specifically, we can observe that the fewer channel numbers and the lower spatial resolution indicate fewer parameters and computation of predictor in step 2. Therefore, benefiting from a fine sampling of analysis filter banks, the computation cost of the earlier scale/mean predictor is negligible. Meanwhile, the multirate channel condition can guarantee parallel decoding and hence effectively speed up the decoding.
- The co-design of the analysis and synthesis filter banks is modular and generic, which can quickly and easily change over from one simple structure to another complex structure to further improve performance without large-scale adjustment of the processing unit.

C. Multi-Level Edge Attention Module

Section III-B indicates that the proposed MPEM is a promising solution for balancing the tradeoff between decoding speed and R-D performance. However, during the spatial sampling process, high-frequency information is lost due to MPEM's multi-rate nature. Entropy coding accuracy suffers due to this information loss at high code rates. Therefore, extracting the more critical and discriminating feature representation for following multirate sampling is still a key challenge in this paper. However, the existing transform structures, such as the attention module, sacrifice inference speed and computational complexity in exchange for a more compact feature representation. They seldom touch on the distinctive importance of texture details and edge information. However, texture details and edge information can be crucial in the scale space of MPEM. Given this, we propose an efficient MEAM to highlight the contribution of the edge and texture information in constructing a more comprehensive scale space and optimize the R-D performance while minimizing any considerable impact on the overall inference speed.

Fig. 4 provides an overview of our MEAM training and inference pipeline. In a nutshell, the edge operator [47] and structural reparameterization principles [54] underpin the design of the proposed MEAM. For a given intermediate feature representation $x \in \mathbb{R}^{C\times H\times W}$, we first construct coarse-grain feature representation $x \in \mathbb{R}^{\frac{C}{2}\times H\times W}$ by a 3×3 standard convolution, and a powerful fine-grained feature representation is further built based on it. Then, inspired by the super-resolution task's success with the edge operator and reparameterization, we create a Multilevel Edge Convolution (MEC) with five distinct functions to better accomplish the above fine-grain feature extraction. More specifically, in **Branch I:** Here, we utilize a standard 3×3 convolution to extract the main feature representations and ensure the basic preference, the convolutional kernel denoted as $w_1 \in \mathbb{R}^{C_{in}\times C_{out}\times 3\times 3}$ where C_{in} is the number of input channels, and C_{out} is the number of the output channel. **Branch II**: Here, we first employ a convolution with a weight of $w_2^1 \in \mathbb{R}^{C\times E\times 1\times 1}$ to expand the channel dimension from C to E ($C < E$), then use the next convolution with a weight of $w_2^2 \in \mathbb{R}^{C\times E\times 3\times 3}$ to squeeze the feature back to C channels. The goal here is to extract wider feature representations, which is confirmed to be able to boost the expressiveness on low-level tasks significantly [55]; **Branch III, IV**: Notice that, unlike previous branches, the core aim of the current branches is to extract the more critical edge or high-frequency information using the pre-defined Sobel filters. It is the 1st-order spatial derivative of feature representations. Specifically, two convolutions with a weight of $w_3^1 \in \mathbb{R}^{C\times C\times 1\times 1}$ and $w_4^1 \in \mathbb{R}^{C\times C\times 1\times 1}$ receive the corresponding original feature representation. Then it was converted and inputted horizontal Sobel filters, and vertical Sobel filters, respectively.

Branch V: Furthermore, removing the non-local maximum in the 1st-order derivatives can detect the finer and more robust edge information. As a result, the Laplacian filter is utilized as the enhanced evolution of 1st-order derivatives on this branch as a typical 2nd-order derivative. Similarly, we also employ convolutions with a weight of $w_5^1 \in \mathbb{R}^{C\times C\times 1\times 1}$ to process the original feature representation, and then the 2nd-order spatial derivative is extracted by a Laplacian filter.

After that, the output feature representation for each branch is added to get an overall MEC output. We re-parameterize the MEC to a standard 3×3 convolution for speedy inference during the inference phase. This process can be presented by:

$$\begin{aligned}\hat{w} &= w_1 + w_2 + w_3 + w_4 + w_5\\ &= w_1 + prem(w_2^1) * w_2^2 + prem(w_3^1) * w_3^2\\ &\quad + prem(w_4^1) * w_4^2 + prem(w_5^1) * w_5^2,\end{aligned} \tag{7}$$

where *perm* is permuting operations to align weights correspondingly. In summary, the MEC leads to feature representation capability promotion using diverse branches and maintains the running time inference of signal 3×3 convolution.

The attention module includes two standard 3×3 convolutions, two MECs, and several nonlinear layers. Notably, two skip connections are used to use feature representation better to speed up convergence. The final output can be obtained as follow:

$$\hat{x} = x + x \odot \delta\left(x'\right), \tag{8}$$

where $\delta(\cdot)$ is the sigmoid function, x' denotes the intermediate feature representations before the sigmoid function.

The MEAM has the following key advantages:

- Due to the MEC's diverse branch structure, the multilevel edge information of the feature representation can be quickly obtained from the pre-built edge operator. Thus, our MEAM can build powerful high-frequency features to recalibrate the feature representation and provide more accurate scale space for following MPEM. Fig. 12 shows that MEC can extract multilevel edge information that is more favorable for image compression tasks through a multi-branch structure.
- Second, our MEAM outperforms the common attention mechanism in LIC in terms of efficiency thanks to the reparameterization mechanism's less complexity and quicker inference during the inference phase.

D. Innovations

To deepen the comprehension of the novelty of the proposed method, we initially distinguish existing efficient entropy models from the proposed MPEM based on the filter banks composition and processing flow, as detailed in Table I. Subsequently, we provide a methodical comparison between the proposed MPEM and each existing method individually. Finally, we conduct a comprehensive analysis of the proposed MEAM and Non-local.

1) Relation to Channel-Based Methods: Based on the single-rate channel sampling of minen2020 [21], Table I illustrates that He2022 [26] and Lu2022 [27] implement multi-rate channel sampling through hand-designed or simple channel ordering rules. Moreover, they collectively utilize various channel context models and spatial contexts to enhance the performance of the processing units. Unlike these methods

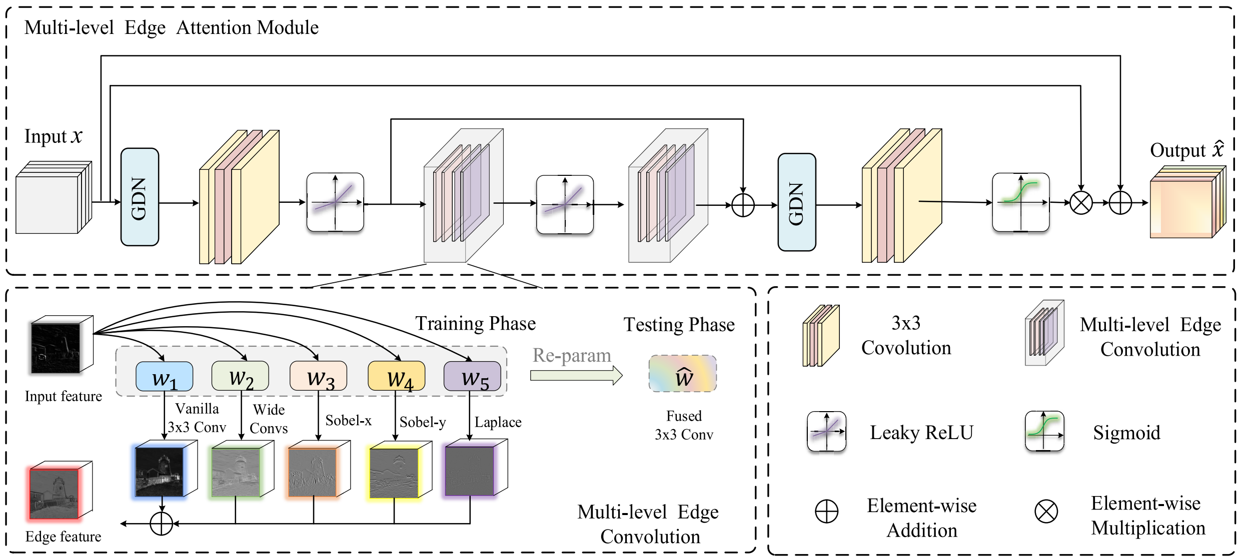


Fig. 4. The detailed flow of our proposed attention structure, wherein the operator weights of different branches, indicated by w_1 to w_5 in the multi-level attention convolution, are distinctly characterized by their shape. Of significance, our proposed methodology reparameterizes these five operators into a single 3×3 convolution during the inference process, thereby rendering it computationally efficient.

relying on basic channel sampling, our proposed MPEM not only samples channels at multiple rates using a meticulously designed set of analysis filter bank but also samples space at corresponding multiple rates. Consequently, with this integrated spatial and channel multirate sampling, we can utilize parallel channel context models and adopt cost-effective spatial predictors within the processing unit, effectively leveraging channel correlation and scale information. This strategy enables MPEM to simplify the processing unit significantly while ensuring precise probability estimation. Ultimately, the design of the synthesized filter bank ensures near-perfect reconstruction of the feature representation, thereby significantly enhancing the overall performance.

2) Relation to Spatial-Based Methods: Our work is also quite different from the existing methods relying on spatial sampling, such as the Hu2020 [22] and the Fu2022 [23]. These methods typically necessitate an additional serial hyper encoder for spatial sampling, thereby compromising parallelism. Moreover, they solely rely on the spatial predictor within the processing unit for probability estimation, disregarding inter-channel correlations, consequently leading to inadequate overall R-D performance. In contrast, our MPEM utilizes a channel-parallel analysis filter bank for multi-rate spatial and channel sampling, enhancing parallelism; it also intelligently integrates the channel context model and spatial predictor within the processing unit. This integration efficiently complements inter-channel correlations, ultimately achieving a superior balance between R-D performance and decoding speed.

3) Relation to Non-Local: Typically, most attention modules in LIC are associated with a non-local operational network to capture potential local and global correlations. However, the nature of non-local operation involves directly capturing distant dependencies by computing interactions between any two locations, leading to a significant computational load. These modules often overlook the unique significance of edge and texture information, which can play a critical role in various low-level tasks. Unlike these methods relying on non-local operation, our proposed MEAM employs multi-branched edge operators to generate powerful high-frequency attention masks during training, somewhat addressing the issue of high-frequency information loss caused by MPEM spatial sampling. In the testing process, the multi-branched structure is consolidated into a single 3×3 convolution through the re-parameterization technique, leading to a reduction in both parameters and computations.

IV. Experiments

A. Datasets and Implementation Details

1) Datasets: We use the CLIC training dataset [56] for training and randomly crop them into 247576 sample images with the shape of $3 \times 256 \times 256$. At the stage of testing, we test our model on the common Kodak dataset with 24 images that has a size of $3 \times 512 \times 768$ or $3 \times 768 \times 512$ and the Tecnick dataset with 100 images of $3 \times 1200 \times 1200$ pixels. We compare our model's R-D performance to that of certain baseline models using the bit-per-pixel (bpp), peak signal-to-noise ratio (PSNR), and multi-scale structural similarity (MS-SSIM). Meanwhile, we also give a comparison of encoding and decoding latency.

2) Implementation Details: The standard CompressAI framework [57] was used to put our model into action. During training, we train six models from scratch to represent six image quality levels, each corresponding to a preset λ. Specifically, we train our models within the ranges $\lambda \in \{0.0018; 0.0035; 0.0067; 0.0130; 0.025; 0.0483\}$, all to optimize the PSNR, with regards to MS-SSIM, $\lambda \in \{0.02; 0.04; 0.07; 0.20; 0.40; 0.605\}$. All models are trained for 100 epochs with a batch size of 16 using the Adam optimizer. The initial learning rate is set to 10e-4, halved at 60, and diminished to 10e-5 at 80. Similar to the setup in [21] and [26], the channel number of our model N; M is set to 192;

320 for all models. The gradient clipping threshold is set to 1 for regular training.

B. Rate-Distortion Performance

To evaluate our method's effectiveness, we first analyzed our model by determining its average R-D performance in terms of PSNR and MS-SSIM on two datasets. In the rate-distortion performance experiment, the Balle2018 [11] model with MEAM and the Bao2022 [43] model, referred to as our model and our deep model, respectively, are used as the main transformation networks. Notably, the nonlinear transformation component within the Bao2022 main transformation resembles the attention mechanism. As a result, we chose not to incorporate MEAM into Bao2022 to guarantee an equitable evaluation. Ablation experiments will be used to demonstrate the efficacy of MEAM. MPEM is used instead of entropy coding in these models, and the group number is set to three. We compare numerous well-known LIC methods, including Minnen2018 [14], Cheng2020 [34], Minnen2020 [21], Hu2020 [22], He2022 [26], Zuo2022 [46], Bao2022 [43], and so on [6], [17], [23], [38], [51], in addition to traditional compression standards like BPG and VVC. Considering that our method and most of the comparison methods are implemented using compreeAI, we have used the VVC results[1] provided by compreeAI as the anchor results for the kodat dataset to ensure fairness in the comparison. However, in the case of the Tecnick dataset, we have utilized our own VTM-10.1[2] test results (YUV444) as the anchor results, given that compressAI did not offer results for the Tecnick dataset. We utilized the script from CompressAI with its default configuration for model evaluation.

Fig. 5 depicts the overall performance comparison results. The Kodak results in Fig. 5(a) show that He2022 achieves the best results in PSNR, and ours-deep achieves a considerable BD-rate -3.35% and outperforms Bao2022 (-1.99%) and VVC in PSNR. Besides, our simple method (ours) performs better than Minne2020 and still achieves the same performance as Cheng2020 at high bit rates. The ours-deep method has proven superior to He2022 and achieves a state-of-the-art BD rate among all the methods optimized for MS-SSIM. The clever use of multiscale information for asymptotic probability estimation in MPEM, which is necessary for calculating MS-SSIM metrics, could be the reason.

Moreover, our straightforward method's MS-SSIM calculation exhibited the same trend as Cheng2020, indicating its efficacy. Furthermore, In light of the Tecnick dataset, the ours-deep model exhibits a notable improvement of -53.03% in terms of rates saved compared to VVC when utilizing MS-SSIM to optimize the model. This outcome corroborates our method's robustness when processing high-resolution images.

C. Encoding and Decoding Cost Analysis

In this section, we present the results of our experimental evaluation of the proposed compression approach's efficacy using the Kodak and Tecnick datasets. Specifically, our evaluation focuses on assessing the efficiency of different methods based on a range of performance metrics, including the average running time, model size, and several parameters. The running time is measured by four different encoding and decoding times, indicating each method's time complexity. We also consider each approach's spatial complexity by evaluating the number of parameters and the actual model size. Unless otherwise specified, all metrics are measured at high bit rates to ensure a fair comparison. Specifically, these comparison methods include Minnen2018 [14], Minnen2020 [21], Cheng2020 [34], He2022 [26], Zuo2022 [46], and Bao2022 [43]. Furthermore, we have introduced several additional comparison methods built upon the same transformation. For instance, by utilizing Bao2022 as the transformation model and integrating entropy models such as He2022 and He2021 [25]. We implemented our own version of He2022 since the official implementation is not available as open source.

We followed a common protocol to test the time with GPU synchronization. To warm up the device, we initially tested the input image (*Kodat19* or *Tecnick01*) 10 times. After the warm-up phase, we averaged the encoding and decoding times over 20 additional tests. Although the average of the 20 test times is reported, each individual round test's average time may have a slight margin of error due to uncontrollable factors such as the machine's operating state.

Table II indicates the indications mentioned above. It shows that the experimental results of the proposed method's fast encoding and decoding capability agreed with the original intention of our highly efficient model design. The proposed simple version method (the first line in kodak&V100) can considerably speed up total decoded runtime compared with some efficient entropy models while achieving a comparable or even better R-D rate in PSNR (i.e., a total decoding time savings of 35.96% compared with that in Minnen20) on the Kodak dataset. Meanwhile, in comparison with the ELIC entropy model in a similar transformation process (e.g., the second line in kodak&V100), our method demonstrates a 61.2% reduction in entropy decoding time, which further attests to the effectiveness of MPEM. The proposed method based on Bao2022's transformation outperforms Bao2022 MS-SSIM RD rate (-50.024% vs. -48.334%). Compared to ELIC with He2022 as the transformation process, our approach demonstrates a minor edge in total decoding time. Notably, this outcome primarily arises from the increased complexity of the transformation in Bao2022 in contrast to He2022, resulting in a comparatively greater portion of the total decoding time attributed to the transformation. Furthermore, when contrasted with ELIC utilizing Bao2022 as the transformation process, our method exhibits a substantial improvement in total encoding speed. Importantly, this improvement is achieved while maintaining a slightly reduced rate-distortion performance, which can be attributed to the effectiveness of our MPEM. Additionally, the calculations and implementation of the proposed method are relatively straightforward (i.e., a spatial complexity savings of 59.1% and 54.8% compared with that in Bao2022 and Zuo2022).

[1]https://github.com/InterDigitalInc/CompressAI/blob/v1.1.9/results/kodak.

[2]https://vcgit.hhi.fraunhofer.de/jvet/VVCSoftware_VTM/-/tags/VTM-10.1

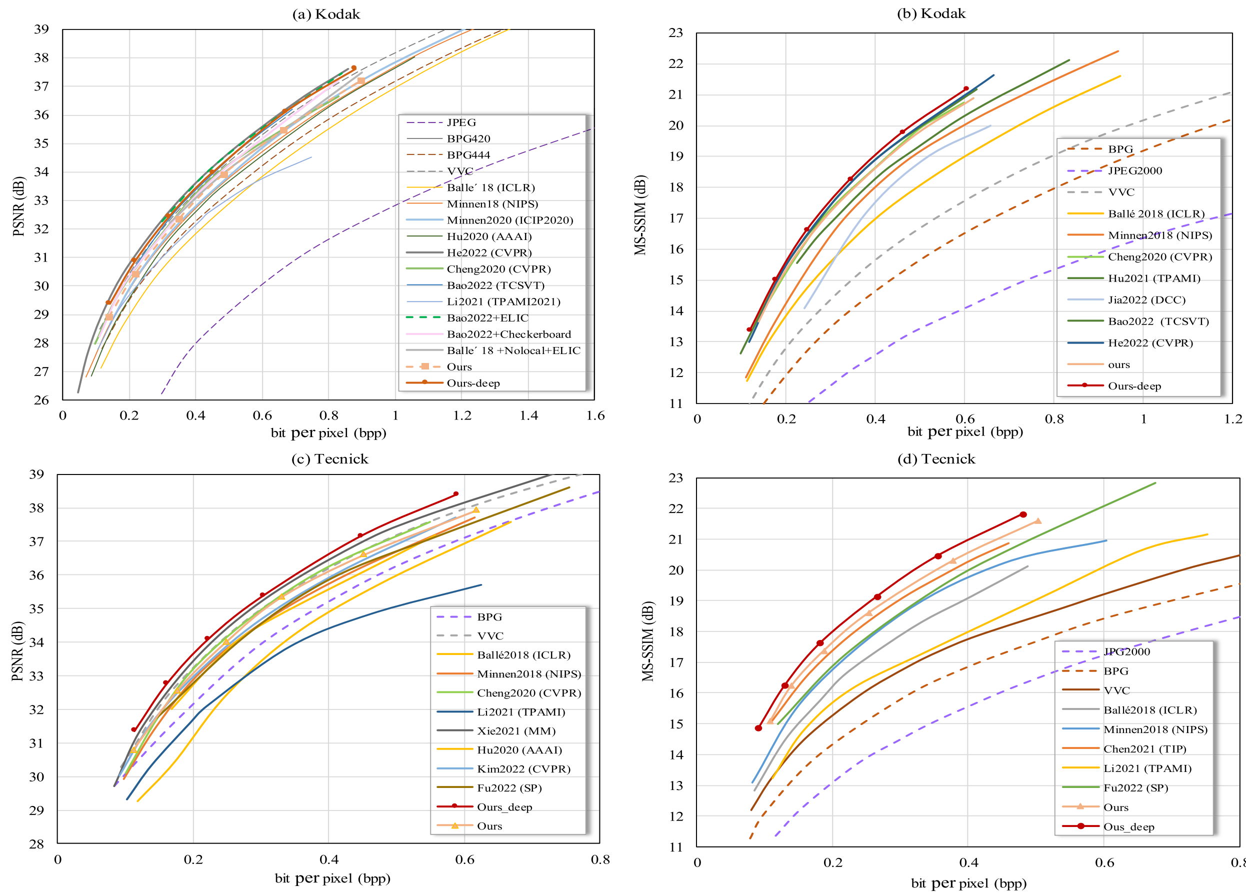


Fig. 5. Comparisons of RD curves with other methods on the Kodak and Tecnick datasets. For a clear illustration, the MS-SSIM values are converted into decibels $-10\log_{10}(1-d)$, where d refers to the MS-SSIM value. "ours" represents the architectural design illustrated in Fig. 1. In order to achieve a just comparison across similar the numbers of parameters and substantiate the universality of the proposed method, we amalgamate the proposed entropy model with the Bao2022 main transformation to construct the "ours-deep" architecture.

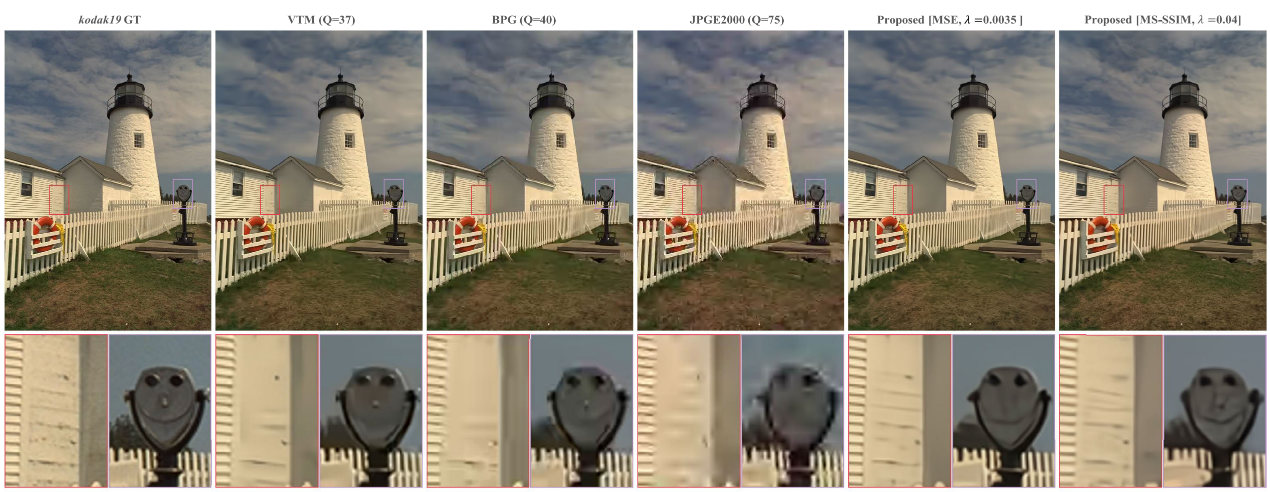


Fig. 6. Visual comparison of reconstructed images at approximately 0.17 bpp.

Except for the Kodak dataset, the experimental result on the Tecnick dataset (V100 and 1080Ti) also represented the same trend. When considering RD-rate, time, and space complexity, this supports that the proposed approach results in effective learned image compression.

D. Qualitative Results Analysis

To assess whether the proposed method can generate more visually pleasant reconstructed images, we performed a visualization comparison of reconstructed images on the Kodak dataset. We compared the proposed method with JPEG2000, BPG, and VVC. We use both MSE and MS-SSIM optimized methods for comparison. The classical image compression standards are processed using CompressAI. Initially, the original image is fed into the relevant model or standard to obtain the reconstructed image. Finally, MulimgView[3] was used to

[3]https://github.com/nachifur/MulimgViewer.

TABLE II

VARIOUS METRICS WERE USED TO COMPARE THE EFFECTIVENESS OF SEVERAL LIC METHODS, INCLUDING THE NUMBER OF PARAMETERS, MODEL SIZE, ENCODING/DECODING TIME. THE TOTAL NETWORK LATENCY FOR ENCODING AND DECODING IS ENCT AND DECT, RESPECTIVELY. WHILE ENCY AND DECY ONLY REFER TO THE LATENCY FOR ENTROPY MODEL ENCODING AND DECODING, RESPECTIVELY

Dateset&GPU	**Methods**		**Efficiency indicators**					
	Transformation	**Entropy model**	**Spatial complexity**		**Running time (ms)**			
			Params↓	Size↓	EncT↓	DecT↓	EncY↓	DecY↓
Kodak&V100 (512 × 764)	Ballé2018 + MEAM	MPEM (proposed)	26.9M	315.3M	72.0	115.9	50.8	41.1
	Ballé2018 + Non-local	ELIC	36.6M	379.1M	139.2	183.0	114.3	105.8
	Ballé2018 [11]	Channel-wise [21]	52.6M	783.2M	117.2	181.0	100.5	111.2
	Ballé2018 [11]	Context + Hyperprior	25.2M	311.5M	3122.7	7811.8	3107.9	7744.6
	Cheng2020 [34]	Context + Hyperprior	29.6M	357.8M	3061.1	7763.1	3024.8	7713.3
	Bao2022 [43]	MPEM (proposed)	34.0M	389.9M	143.7	191.1	48.4	41.3
	Bao2022 [43]	ELIC	45.6M	530.6M	208.2	254.5	111.7	104.7
	Bao2022 [43]	Checkerboard	51.4M	608.2M	177.9	207.8	80.8	56.9
	Bao2022 [43]	Channel-wise [21]	83.1M	908.7M	197.8	259.4	101.7	108.8
	Zuo2022 [46]	Channel-wise [21]	75.2M	861.6M	128.5	187.3	102.4	105.2
	He2022 [26]	ELIC	38.5M	449.2M	144.7	194.6	112.1	107.4
	He2022 [26]	MPEM (proposed)	26.93M	315.9M	81.8	128.0	49.0	41.1
Tecnick &V100	Bao2022 [43]	MPEM (proposed)	34.0M	389.9M	480.5	701.2	162.9	136.0
		ELIC	45.6M	530.6M	602.3	801.9	282.6	233.2
		Checkerboard	51.4M	608.2M	553.1	751.4	232.4	182.1
Tecnick &1080Ti	Bao2022 [43]	MPEM (proposed)	34.0M	389.9M	794.4	1001.7	176.5	149.3
		ELIC	45.6M	530.6M	939.7	1083.2	321.1	231.0
		Checkerboard	51.4M	608.2M	887.3	1038.0	270.8	185.2

Fig. 7. Visual comparison of reconstructed images at approximately 0.1 bpp.

select and zoom in on specific areas of the reconstructed map. In Fig 6, with a bpp approximation of 0.17, both methods of "ours-deep" utilize a model of the Q2 (MSE:$\lambda = 0.0035$, MS-SSIM:$\lambda = 0.04$). To determine the approximate Bpp, VTM is tested with Q of 37, JPEG2000[4] with Q of 75, and BPG[5] with Q of 40. As seen in Fig. 6, our method can capture more texture information, such as the door and the smiling face, compared with other methods. The JPEG2000 achieves poor performance owing to the ringing artifacts. The BPG codec achieves a smooth result, but some regions do not preserve the details and delicate structures.

For reconstructed maps with a bpp close to 0.1, all instances of "ours-deep" are modeled using Q1(MSE:$\lambda = 0.0018$, MS-SSIM:$\lambda = 0.024$), while the classical method is configured as shown in Fig 7. As depicted in Fig 7, contrasted with

[4]https://ffmpeg.org/

[5]https://bellard.org/bpg/libbpg-0.9.8.tar.gz

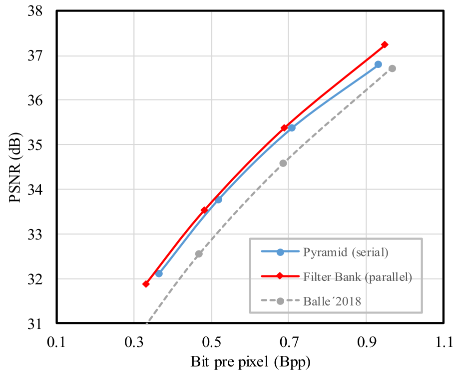


Fig. 8. Ablation experiments of feature processing order in MEPM, where pyramids are processed serially, and filter banks are processed in parallel.

the visual results at 0.17bpp, the visual results of all the reconstructed maps, approximating 0.1 bpp, exhibit a certain degree of degradation. Nevertheless, our MS-SSIM-based optimization method consistently attains superior visual results, notably emphasizing the regions of the lighthouse wall and smiley face. Conversely, the BPG demonstrates unpleasant reconstruction results, particularly noticeable on the right side of the lighthouse. In conclusion, our method can obtain better visual quality than other methods.

V. Ablation Studies

To validate the effect of each component in our method, we further experimented with several ablation experiments with the leave-one-out method and other standard methods. Notably, the main transformation network for all ablation experiments is ICLR2018 [11], where the model is replaced by ours MPEM, and MEAM is available as an optional add-on module. We use ICLR2018 as the anchor in the ablation experiment in contrast to the performance experiment to make the B-D rate more apparent. The standard Kodak dataset served as the basis for every ablation experiment. The results are summarized in the following sections:

A. Multirate Progressive Entropy Model

1) Filter Banks Vs. Pyramid: Filter Banks, which is essentially a form of parallel processing, is also an efficient subband coding. The pyramid method is another way to save computing power. Unlike filter banks, which are serial processes, the pyramid method is not sequential. As a result, to take into account the effect that the feature processing order has on the proposed MPEM, we first evaluate the MPEM's performance using a variety of branch structure combinations, like a three-layer pyramid and a three-layer filter banks. Fig 8 shows that filter banks and pyramid structures outperformed the baseline regarding R-D performance. Specifically, when the filter banks is used as the feature processing order, our method saves considerable RD rates. In contrast, only an less RD-rate saving is obtained using the pyramid structure, proving that the serial feature processing order is more suitable for our entropy coding model. The fact that the pyramid structure sequentially builds the scale space using the same feature representation is one possibility. This unavoidably leads to redundancy in

TABLE III

Ablation Study About Different Dimensional Sampling Methods

Methods	Sampling		BD-Rates↓	Flops.	
	Channel	Spatial		Latencys ↓	
MPEM	*single-rate*	*single-rate*	-0.083	80.49G	
	single-rate	*multi-rate*	-9.218	93.98G	
	multi-rate	*single-rate*	2.863	75.93G	-
	multi-rate	*multi-rate*	-9.771	87.97G	

TABLE IV

Different Designs of Channel Sampling Rate and Spatial Sampling Rate in MPEM

Methods	Groups	Channel Sampling	Spatial Sampleing
MPEM	2	$[32, 288]$	$[8 \times 8, 16 \times 16]$
	3	$[32, 32, 256]$	$[4 \times 4, 8 \times 8, 16 \times 16]$
	4	[32, 32,64, 192]	$[2 \times 2, 4 \times 4, 8 \times 8, 16 \times 16]$

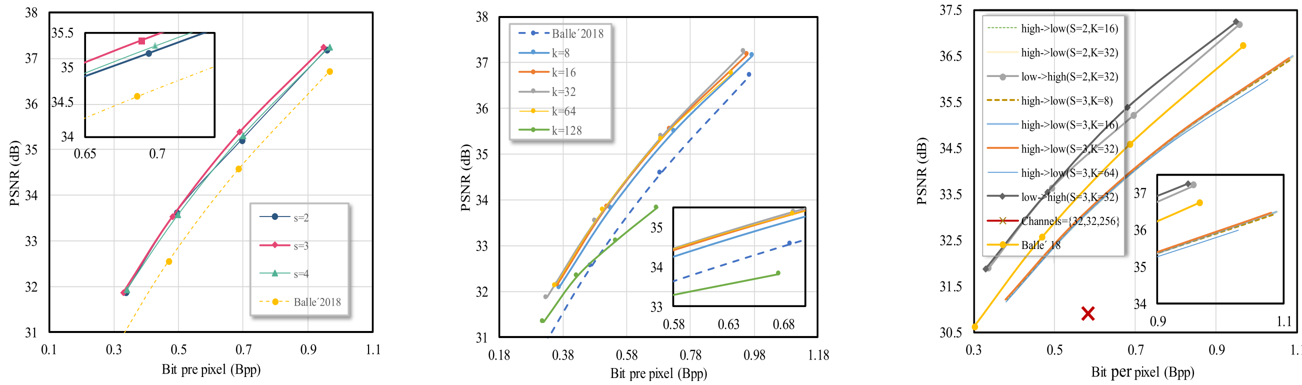
the feature representation and taints the accuracy of entropy coding.

2) Multi-Rate Vs. Single-Rate: After confirming that filter banks can positively affect our entropy model, our next focus was to investigate the impact of the multiple and single-rate sample methods on the R-D performance and Flops Latency of the proposed MPEM by an experimental design.

We use a 4-group MPEM as the experimental object because it can be divided by the entropy model's channel number $M = 320$. Multirate channel sampling means that the channel numbers in each group are sampled as {32, 32, 64, 192}. This also conforms to the energy compaction characteristics proposed in [26]. Moreover, multirate spatial sampling means that each group is compactly sampled as $\{2 \times 2, 4 \times 4, 8 \times 8, 16 \times 16\}$. The opposite is single-rate sampling, in which the spatial resolution is uniformly sampled as 8×8, and each group is uniformly sampled as the original channel number quarter. As listed in Table III, in comparison to single-rate sampling techniques employed for both channel and spatial dimensions, multi-rate sampling for the chanel dimension further reduces Flops. However, this approach may result in a slight decrease in the B-D rate.

BD-rate is significantly improved and significantly reduced when multi-rate sampling is applied to both the spatial and channel dimensions. This is because of the LIC's channel energy concentration, which means that the more forward channels have more energy. In contrast, spatial single-rate sampling performs the same operation for all channels, leading to decreased performance. The channel energy characteristics can be effectively matched using spatial multi-rate sampling, resulting in a reasonable sampling rate distribution. This reduces Flops.

3) Number of Groups in the Filter Bank: The above multi-rate sampling is the core operation in our MPEM. Therefore the shape of its dimensional space (e.g., number and size) also greatly affects the performance of the entropy model. To this end, we conducted detailed ablation experiments on constructing dimensional space. Table IV depicts the main settings of



(a) Number of groups in the filter banks (b) Number of channel-wise groups (c) Encoding order of scale space

Fig. 9. Ablation experiments with different channel, spatial sampling rates and scale space's encoding order in MPEM.

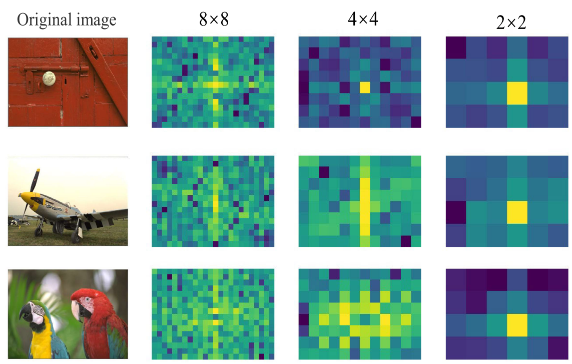


Fig. 10. Visualization of the amplitude map of the feature spectrum. It can be seen that the strong signal in the feature map after the upsampling operation is repeated at specific frequencies.

these experiments, and Fig. 9(a) depicts the outcomes. It can be seen that the model has the best RD performance when the number of groups is set to three; The network's performance shows a slight decrease when the number of groups is set to 4. This may happen because the model over-samples the spatial dimension as the number of groups increases (for example, the first group's feature representation's spatial resolution is sampled as 2×2), which corrupts the feature representation to some extent and prevents it from being used for accurate entropy coding with the full decomposition model. A similar trend is also observed when the number of groups is limited to only two ($s = 2$), which suggests that the insufficient number of groups impedes MPEM from utilizing adequate multiscale information to estimate probabilities accurately.

4) Number of Channel-Wise Groups in the Filter Banks: In this part, we select the 3-groups entropy model in Table IV as the present experimental baseline by observing the experimental results in Fig. 9(b). Like [26], we set channel numbers at the first two groups as k and channel numbers of the remaining group as $m-2k$. However, these ablation techniques experience a decrease in PSNR and bpp performance due to the variation in channel numbers among groups in the same baseline. For instance, the model's R-D rate decreases by 1.515% when K is set to 16. The fact that a smaller K value indicates less information about small-scale features is the primary cause of this phenomenon. Hence, the entropy model cannot fully use small-scale information to construct a complete scale space. The R-D performance increased with the increase of the K value, which had a considerable gain after K increased to 32. However, when K was set to 64, the RD performance of the model showed a slight decrease.

Moreover, when K is increased to 128, the model's performance significantly decreases, even below ICLR18 at a high bit rate. This may be because, even though more channels can improve scale space refinement, feature representations on more channels are also significantly sampled, resulting in the loss of key feature information for entropy coding models at high bit rates. In conclusion, the entropy model balances information sampling loss and completeness of the scale space for $K = 32$.

5) Progressive Coding: As an important factor of MPEM, progressive coding has contributed significantly to the accuracy of probability estimation. In the meantime, the overall entropy estimation's performance is significantly influenced by the scale space's encoding order. In RSNR and bpp, hight→low [32,32,256] performs worse than low→hight, as shown in Fig. 9(c).

From the experimental setup of the model mentioned above, it appears that the downsampling of the 256 channels to 4×4 results in the loss of a substantial amount of spatial information during the sampling process. Consequently, the high-frequency portion of the 32 channels cannot be effectively leveraged to precisely estimate the probability, leading to the model's slow convergence and its descent into a local optimum point. To validate this assertion, we maintained the 256 channels in a 16×16 spatial size, with 32 channels sampled at 8×8, and another group sampled at 4×4. The results of the experiments are illustrated in Fig. R4, indicating that compared to the earlier configuration [32,32,256], the "low→high" configuration shows significantly improved R-D Performance. Although the "low→high" model successfully converges to a more appropriate code rate range, it is still lower than the "high→low" model.

To determine the optimal number of channel groupings for the "high-low" model, we conducted an ablation experiment on the channel grouping count in the "high-low" model. We specifically configured the channel grouping number as [M-2K, K, K]. The figure illustrates that for $K = 64$, the R-D

TABLE V

MPEM Encompasses Various Configurations of Its Components, Which Also Include w/o Attention (Non-Local and MEAM), g_{cc} and g_{sm} Are Described in Section III-B, and Post-Processing Indicates the Last Three Resblocks

Methods	$g_{cc}(\cdot)$	$g_{sm}(\cdot)$	Post-processing	Attention
Configuration 1	✔	✗	✗	✗
Configuration 2	✔	✔	✗	✗
Configuration 3	✔	✗	✔	✗
Configuration 4	✔	✔	✔	✗
Configuration 5	✔	✔	✔	Non-local
Configuration 6	✔	✔	✔	MEAM

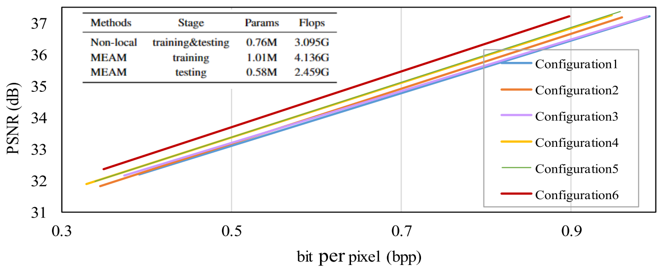


Fig. 11. Ablation study about the performance contribution of different components in MPEM.

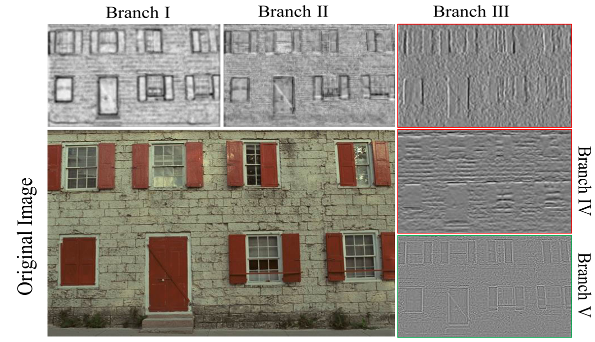


Fig. 12. The feature visualization of different branches in MEC, from which it can be seen that MEC can extract more useful edge and texture information for LIC through different branches.

curve of the model diminishes at the high bit rate, suggesting that an increase in spatially sampled channels is detrimental to the probability estimation of the "high→low" model.However, when K is set to 8, the R-D curve of the model decreases slightly. This phenomenon can be attributed to the compression of feature representations from 304 to 16 channels in the first set of spatial predictors, depicted in the lower right part of Fig 2. This compression is utilized for obtaining entropy parameter (μ, σ) in the 8 channels segment of the model, resulting in a loss of a priori information. In the "low-high" modeling process, the spatial predictor functions as a channel expander and spatial upsampler, thereby generating channel and spatial a priori information that is more conducive to probability estimation, as illustrated in Fig. 10. This could also explain the widespread use of similar low→high sequences in other methods [23], [32].

We also show the magnitude map of the feature spectrum in each spatial scale, which represents channel with the highest entropy, to learn more about that process. For clarity, we move low frequencies to the spectrum map's center and high frequencies to the corners of the spectrum map. Fig. 10 provides a visual representation of the results. From the magnitude map, we can perceive that the up-sampling operation in the parameter predictor repeats strong frequency components of the input to generate spectrums of upper layers [53], equivalent to providing more a priori information for the next entropy encoding/decoding. The down-sampling operation, on the other hand, is the upsampling operation reversed, which could also explain why the low→hight ratio is superior to the hight→low.

6) The Specific Configuration of MPEM: To understand the contribution of our MPEM, we compared the RD performance of the different components. The different network configurations of MPEM are shown in Table V, and the corresponding results are shown in Fig. 11.

First, the processing unit's scale/mean predictor can save considerable BD rate when comparing the performance of configurations 1 and 2. Moreover, when carrying out the model with configuration 4, we can obtain further performance gains, which, thanks to the post-processing module, eliminate the distortion due to multirate spatial sampling. In addition, we removed the $g_{sm}(\cdot)$ from Configuration 4 to form configuration 3. The performance under configuration 3 had to reduce to a certain degree when compared to Configuration 4, demonstrating the advantages of this scale/mean predictor.

B. Multi-Level Edge Attention Module

1) MEAM Vs. Non-Local: The high-frequency feature extraction capability will vary with different branches of MEAM (as show in Fig. 12). In addition, as shown in the Fig. 11, configuration 5 demonstrates that common Non-local attention yields only negligible performance improvement on MPEM, while configuration 6 indicates that combining our MEAM with MPEM results in a substantial improvement in R-D performance. This demonstrates that MEAM effectively mitigates the high-frequency information loss limitation of MPEM by incorporating multi-level edge information, making our proposed MEAM more compatible to MPEM.

To evaluate the computational efficiency of our proposed MEAM, we compare it with the Non-local method. We conducted separate attention module tests using an input tensor of dimensions $192 \times 64 \times 64$. As depicted in the Fig. 11, the computational complexity of our MEAM varies between the training and testing phases due to the re-parameterization technique adopted. During the training phase, MEAM exhibits a larger number of parameters and computations than Non-local due to its multi-branching structure. However, during the testing phase, the multi-branching structure of MEAM can be re-parameterized into a single 3×3 convolution (as shown in the bottom left of Fig. 4). As a result, both the computations and the number of parameters are significantly reduced, demonstrating the efficient inference capability of MEAM.

2) The Specific Configuration of MEAM: Our proposed MEAM aims to emphasize the significance of edge and texture

TABLE VI
DIFFERENT CONFIGURATIONS OF PRE-DEFINED OPERATOR BRANCHES IN MEAM

Methods	Base	Sobel-x	Sobel-y	Laplace
Configuration 1	✓	✗	✗	✗
Configuration 2	✓	✓	✗	✗
Configuration 3	✓	✓	✓	✗
Configuration 4	✓	✗	✗	✓
Configuration 5	✓	✓	✓	✓

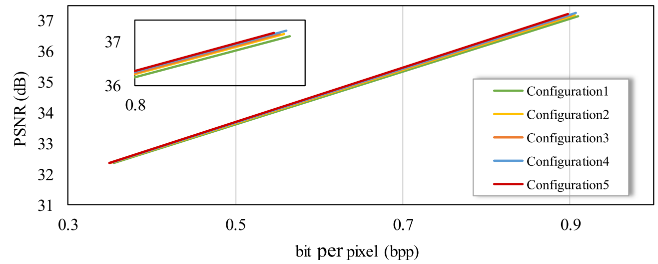


Fig. 13. Ablation study about the performance contribution of different pre-defined operators in MEAM.

information in constructing a more comprehensive scale space and optimizing the R-D performance. The extraction of edge and texture information relies on pre-defined traditional operator branches. Consequently, we have incorporated a series of ablation studies to provide a clearer understanding of the contributions of these pre-defined operators. The corresponding ablation configurations are depicted in Table VI, and the results are illustrated in Fig. 13.

As depicted in Fig. 13, Configuration 2, incorporating Sobel X, exhibits a relatively smaller performance gain at high code rates in contrast to Configuration 1, which excludes the predefined operators. This trend is similarly observed in Configuration 3, which integrates Laplace operators into Configuration 1. The potential explanation for this phenomenon might stem from the limited learning capability of a single branch, despite the Laplace operator being a more stable second-order operator than Sobel X. However, in Configuration 4, the joint utilization of Sobel X and Sobel Y operators enables the further augmentation of edge and texture information in both directions, leading to a more significant performance enhancement at higher bit rates. The optimal performance of MPEM is achieved when the three edge operators are jointly employed. Of noteworthy mention is the near absence of performance disparities among all the compared methods at low bit rates. This phenomenon can potentially be attributed to the lesser demand for edge and texture in the model at low bit rates. In contrast, at high bit rates, the reconstruction process heavily relies on ample texture and edge information to ensure the quality of the reconstruction.

3) Visualization for Channel Energy: In LIC, channel energy is an important index to assess the feature conversion capability of the main transformation network. Thus, we follow [43] to statistically analyze the energy distribution change in the different channels before and after the attention transform. For fairness, we use Cheng2020 [34] as the comparison standard and select the features before and following the transformation of the second layer as the visual object. Then, for clear presentation, we select *kodat*12 and *kodat*19 as the input images, sort the energy before and after the transformations, and only display the first sixty channels. The visual results shown in Fig. 14, as can be seen, compared with Cheng2020, the MPEM transformed energy is more concentrated (0.601 v.s. 0.719)), and at the same time, MPEM's pre- and post-transformation larger differences show that MEAM performs better at energy concentration than non-local, and compact energy improves compression performance [12], [43]. That can explain why MEAM is more efficient.

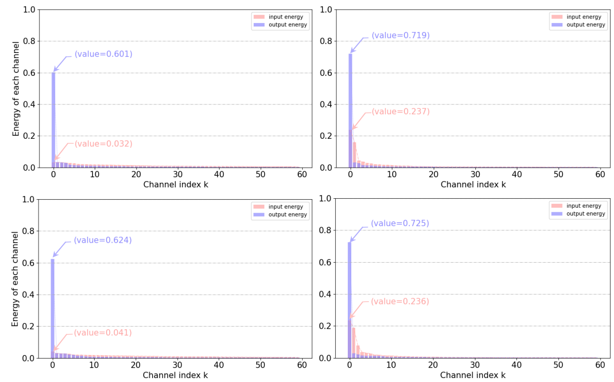


Fig. 14. Visualization of different attention methods using *kodat*17 and *kodat*9. Higher ability values and more concentrated distribution indicate a more compact feature representation.

VI. CONCLUSION

This paper approaches LIC efficient entropy model design from a novel angle. We begin by comparing and summarizing the limitations of the existing split-coded-then-merge strategies for efficient entropy coding to filter banks in conventional signal processing based on their similarity to optimization objectives and structure. The fundamental design idea of multirate filter banks inspires our proposal: an effective multirate progressive entropy model. Finally, we propose a multi-level edge attention module for building a complete scale space to reduce the high-frequency information loss caused by MPEM's multi-rate sampling. The experimental outcomes demonstrate the efficacy and generalizability of our method. In future work, we aim to design finer analysis filter banks and synthesis filter banks to achieve the perfect reconstruction of the feature representation.

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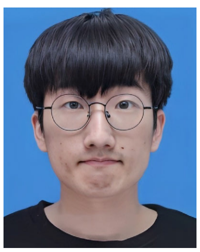

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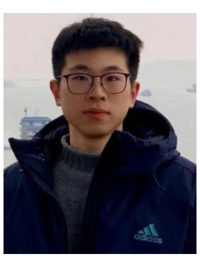

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